Key resources table

The key resources table (KRT) highlights the reagents, genetically modified organisms and strains, cell lines, software, instrumentation, and source data **essential** to reproduce results presented in the manuscript. Depending on the nature of the study, this may include standard laboratory materials (i.e., food chow for metabolism studies, support material for catalysis studies), but the table is **not** meant to be a comprehensive list of all materials and resources used (e.g., essential chemicals such as standard solvents, SDS, sucrose, or standard culture media do not need to be listed in the table). **However, please note that items in the table must also be reported in the method details section within the context of their use.**

ALL references cited in the KRT must be included in the main references list.

- For **Cell Press journals**, citations should be formatted as "Author name et al.[#]" (e.g., Smith et al.[1]), with the citation number matching that in the main references list.
- For ***Stem Cell Reports***, citations should be written as "Author name et al., publication year" (e.g., Smith et al., 2024).

Please report the information as follows:

- **REAGENT or RESOURCE:** provide the full descriptive name of the item so that it can be identified and linked with its description in the manuscript (e.g., provide version number for software, host source for antibody, strain name, etc.). See the sample tables at the end of this document for examples of how to report reagents.
    - In the **experimental models** sections (applicable only to experimental life science studies), please include all models used in the study and describe each line/strain as model organism: name used for strain/line in study: genotype (e.g., Mouse: OXTR$^{fl/fl}$: B6.129(SJL)-Oxtr$^{tm1.1Wsy/J}$).
    - The **biological samples** section (applicable only to experimental life science studies) should list all samples obtained in this study or from commercial sources or biological repositories.
    - **You may list a maximum of 10 oligonucleotides or RNA sequences** in the table. If there are more than 10 to report, please provide this information as a supplemental document and reference the file (e.g., See Table S1 for XX) in the KRT.
    - **Deposited data** should include both newly deposited data from this study and existing datasets that were used in the study.
    - Please include software and code mentioned in the method details and data and code availability sections under **software and algorithms**.
    - Any item that does not fit the existing subheadings should be added to the "other" section. **Please do not add your own subheadings.**
- **SOURCE:** report the company, manufacturer, or individual that provided the item or from where the item can be obtained (e.g., stock center or repository).
    - For materials distributed by Addgene, please cite the article describing the plasmid and include "Addgene" as part of the identifier.

- If an item is from another lab, please include the name of the principal investigator and/or a citation if it has been previously published.
- If the material is being reported for the first time in the current paper, please indicate as "this paper."
- For software, please provide the company name if it is commercially available or cite the paper in which it had been initially described.

- **IDENTIFIER:** include catalog numbers (entered in the column as "Cat#" followed by the number, e.g., Cat#3879S). Where available, please include unique entities such as RRIDs, Model Organism Database numbers, and accession numbers preceded by database abbreviations such as PDB or CCDC). Please ensure the accuracy of the identifiers, as they are essential for generation of hyperlinks to external sources when available. For more information about data sharing policies and a list of recommended data repositories for abbreviations, please see the Cell Press Author's guide to data sharing.
    - For antibodies, if applicable and available, please also include the lot number or clone identity.
    - For software or data resources, please include the URL for where the resource can be downloaded.
    - When listing more than one identifier for the same item, use semicolons to separate them (e.g., Cat#3879S; RRID: AB_2255011).
    - If an identifier is not available, please enter "N/A" in the column.
    - **A NOTE ABOUT RRIDs:** we highly recommend using RRIDs as the identifier (in particular, for antibodies and organisms but also for software tools and databases). For more details on how to obtain or generate an RRID for existing or newly generated resources, please visit the RII or search for RRIDs.

Completing your table

Please use the empty table that follows to organize the information under the provided subheadings and skip sections that are not relevant to your study.

- To add a row, place the cursor at the end of the row above where you would like to add the row, just outside the right border of the table. Then press the ENTER key to add the row. Alternatively, you can right-click on your mouse and choose Insert > Insert rows above or Insert rows below.
- Please delete empty rows (right-click on your mouse and select Delete rows or Delete cells).
- Each entry must be on a separate row; do not list multiple items in a single table cell.

Please see the appendix A at the end of this document for examples of how reagents and instrumentation should be cited in the life and physical sciences.

TABLE FOR AUTHOR TO COMPLETE

Please do not add custom subheadings. If you wish to make an entry that does not fall into one of the subheadings below, please contact your handling editor or add it under the "other" subheading. **Any subheadings not relevant to your study can be skipped.** (**NOTE:** references should be in numbered style, e.g., Smith et al.[1], for Cell Press journals and Harvard style, e.g., Smith et al., 2024, for *Stem Cell Reports*.)

Key resources table

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Bacterial and virus strains		
Venezuelan equine encephalitis virus (VEEV), attenuated vaccine strain, TC-83Venezuelan equine encephalitis virus (VEEV), vaccine strain TC-83	I. FrolovP1 laboratory stock generated from an infectious cDNA clone provided by I. Frolov	Kinney, JVI 1992Kinney et al., J. Virol. 1993
Chemicals, peptides, and recombinant proteins		
Dulbecco's phosphate-buffered saline (DPBS)	Corning	Cat#21-030-CV
EGM-2 Endothelial Cell Growth Medium-2 BulletKit	Lonza	Cat#CC-3162
Fetal bovine serum (FBS)	Gibco	Cat#A5256801
Dulbecco's Modified Eagle Medium (DMEM)	Quality Biological	Cat#112-014-101CS
Trypsin (0.25%)-EDTA (0.02%)	Quality Biological	Cat#118-093-721
Minimum Essential Medium (EMEM) (2X) without Phenol Red and L-Glutamine	Quality Biological	Cat#115-073-101
Penicillin/Streptomycin solution (100X)	Corning	Cat#30-002-CI
Experimental models: Cell lines		
Primary human brain microvascular endothelial cellsHuman: Primary brain microvascular endothelial cells (HBMVEC), male pediatric donor, passages 5-8	Cell SystemsCell Systems (an AnaBios company)	ACBRI 376Cat#ACBRI 376; Lot#376.07.05.01.2F
Software and algorithms		
CELLCYTE StudioCELLCYTE Studio	CELLCYTE SoftwareCYTENA	www.cytena.comhttps://www.cytena.com/products/cellcyte-x/
GraphPad Prism v11.0.0GraphPad Prism v11.0.0	GraphPad SoftwareGraphPad Software	www.graphpad.com https://www.graphpad.com
Python (v3.10)	Python Software Foundation	https://www.python.org
PyTorch (v2.0)	PyTorch	https://pytorch.org
torchvision (ResNet50 with ImageNet pretrained weights)	PyTorch	https://pytorch.org/vision

ResNet50 architecture	He et al.19	https://pytorch.org/vision/stable/models/resnet.html
scikit-learn	Pedregosa et al.	https://scikit-learn.org
t-SNE (visualization)	van der Maaten and Hinton20	https://scikit-learn.org/stable/modules/generated/sklearn.manifold.TSNE.html
cell_tempo analysis pipeline	This paper	https://github.com/CAIR-LAB-WFUSM/cell_tempo
Other		
Dulbecco's phosphate-buffered saline (DPBS)CELLCYTE X live-cell imaging system (10x objective, Enhanced Contour mode)	CorningCYTENA GmbH, Freiburg, Germany	21-030-CVCELLCYTE X Quick Start Guide; https://www.cytena.com/products/cellcyte-x/
EGM®-2 Endothelial Cell Growth Medium-2 BulletKit®12-well tissue-culture plates (CELLSTAR)	LonzaGreiner Bio-One	CC-3162Cat#665180

Appendix A. Sample tables for author reference

LIFE SCIENCES

REAGENT or RESOURCE	**SOURCE**	**IDENTIFIER**
Antibodies		
Rabbit monoclonal anti-Snail	Cell Signaling Technology	Cat#3879S; RRID: AB_2255011
Mouse monoclonal anti-Tubulin (clone DM1A)	Sigma-Aldrich	Cat#T9026; RRID: AB_477593
Rabbit polyclonal anti-BMAL1	This paper	N/A
Bacterial and virus strains		
pAAV-hSyn-DIO-hM3D(Gq)-mCherry	Krashes et al.[1]	Addgene AAV5; 44361-AAV5
AAV5-EF1a-DIO-hChR2(H134R)-EYFP	Hope Center Viral Vectors Core	N/A
Cowpox virus Brighton Red	BEI Resources	NR-88
Zika-SMGC-1, GENBANK: KX266255	Isolated from patient (Wang et al.[2])	N/A
Staphylococcus aureus	ATCC	ATCC 29213
Streptococcus pyogenes: M1 serotype strain: strain SF370; M1 GAS	ATCC	ATCC 700294
Biological samples		

Healthy adult BA9 brain tissue	University of Maryland Brain & Tissue Bank; http://medschool.umaryland.edu/btbank/	Cat#UMB1455
Human hippocampal brain blocks	New York Brain Bank	http://nybb.hs.columbia.edu/
Patient-derived xenografts (PDX)	Children's Oncology Group Cell Culture and Xenograft Repository	http://cogcell.org/
Chemicals, peptides, and recombinant proteins		
MK-2206 AKT inhibitor	Selleck Chemicals	S1078; CAS: 1032350-13-2
SB-505124	Sigma-Aldrich	S4696; CAS: 694433-59-5 (free base)
Picrotoxin	Sigma-Aldrich	P1675; CAS: 124-87-8
Human TGF-β	R&D	240-B; GenPept: P01137
Activated S6K1	Millipore	Cat#14-486
GST-BMAL1	Novus	Cat#H00000406-P01
Critical commercial assays		
EasyTag EXPRESS 35S Protein Labeling Kit	PerkinElmer	NEG772014MC
CaspaseGlo 3/7	Promega	G8090
TruSeq ChIP Sample Prep Kit	Illumina	IP-202-1012
Deposited data		
Raw and analyzed data	This paper	GEO: GSE63473
B-RAF RBD (apo) structure	This paper	PDB: 5J17
Human reference genome NCBI build 37, GRCh37	Genome Reference Consortium	http://www.ncbi.nlm.nih.gov/projects/genome/assembly/grc/human/
Nanog STILT inference	This paper; Mendeley Data	http://dx.doi.org/10.17632/wx6s4mj7s8.2
Affinity-based mass spectrometry performed with 57 genes	This paper; Mendeley Data	Table S8; http://dx.doi.org/10.17632/5hvpvspw82.1
Experimental models: Cell lines		
Hamster: CHO cells	ATCC	CRL-11268
D. melanogaster: Cell line S2: S2-DRSC	Laboratory of Norbert Perrimon	FlyBase: FBtc0000181
Human: Passage 40 H9 ES cells	MSKCC stem cell core facility	N/A

Human: HUES 8 hESC line (NIH approval number NIHhESC-09-0021)	HSCI iPS Core	hES Cell Line: HUES-8
Experimental models: Organisms/strains		
C. elegans: Strain BC4011: srl-1(s2500) II; dpy-18(e364) III; unc-46(e177)rol-3(s1040) V.	Caenorhabditis Genetics Center	WB Strain: BC4011; WormBase: WBVar00241916
D. melanogaster: RNAi of Sxl: y[1] sc[*] v[1]; P{TRiP.HMS00609}attP2	Bloomington Drosophila Stock Center	BDSC:34393; FlyBase: FBtp0064874
S. cerevisiae: Strain background: W303	ATCC	ATTC: 208353
Mouse: R6/2: B6CBA-Tg(HDexon1)62Gpb/3J	The Jackson Laboratory	JAX: 006494
Mouse: OXTRfl/fl: B6.129(SJL)-Oxtr$^{tm1.1Wsy}$/J	The Jackson Laboratory	RRID: IMSR_JAX:008471
Zebrafish: Tg(Shha:GFP)t10: t10Tg	Neumann and Nuesslein-Volhard[3]	ZFIN: ZDB-GENO-060207-1
Arabidopsis: 35S::PIF4-YFP, BZR1-CFP	Wang et al.[4]	N/A
Arabidopsis: JYB1021.2: pS24(AT5G58010)::cS24:GFP(-G):NOS #1	NASC	NASC ID: N70450
Oligonucleotides		
siRNA targeting sequence: PIP5K I alpha #1: ACACAGUACUCAGUUGAUA	This paper	N/A
Primers for XX, see Table SX	This paper	N/A
Primer: GFP/YFP/CFP Forward: GCACGACTTCTTCAAGTCCGCCATGCC	This paper	N/A
Morpholino: MO-pax2a GGTCTGCTTTGCAGTGAATATCCAT	Gene Tools	ZFIN: ZDB-MRPHLNO-061106-5
ACTB (hs01060665_g1)	Life Technologies	Cat#4331182
RNA sequence: hnRNPA1_ligand: UAGGGACUUAGGGUUCUCUCUAGGGACUUAGGGUUCUCUCUAGGGA	This paper	N/A
Recombinant DNA		
pLVX-Tight-Puro (TetOn)	Clonetech	Cat#632162
Plasmid: GFP-Nito	This paper	N/A
cDNA GH111110	Drosophila Genomics Resource Center	DGRC:5666; FlyBase:FBcl0130415
AAV2/1-hsyn-GCaMP6- WPRE	Chen et al.[5]	N/A
Mouse raptor: pLKO mouse shRNA 1 raptor	Thoreen et al.[6]	Addgene Plasmid #21339
Software and algorithms		
ImageJ	Schneider et al.[7]	https://imagej.nih.gov/ij/

Bowtie2	Langmead and Salzberg[8]	http://bowtie-bio.sourceforge.net/bowtie2/index.shtml
Samtools	Li et al.[9]	http://samtools.sourceforge.net/
Weighted Maximal Information Component Analysis v0.9	Rau et al.[10]	https://github.com/ChristophRau/wMICA
ICS algorithm	This paper; Mendeley Data	http://dx.doi.org/10.17632/5hvpvspw82.1
Other		
Sequence data, analyses, and resources related to the ultra-deep sequencing of the AML31 tumor, relapse, and matched normal	This paper	http://aml31.genome.wustl.edu
Resource website for the AML31 publication	This paper	https://github.com/chrisamiller/aml31SuppSite

PHYSICAL SCIENCES

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Chemicals, peptides, and recombinant proteins		
QD605 streptavidin conjugated quantum dot	Thermo Fisher Scientific	Cat#Q10101MP
Platinum black	Sigma-Aldrich	Cat#205915
Sodium formate BioUltra, ≥99.0% (NT)	Sigma-Aldrich	Cat#71359
Chloramphenicol	Sigma-Aldrich	Cat#C0378
Carbon dioxide (^{13}C, 99%) (<2% ^{18}O)	Cambridge Isotope Laboratories	CLM-185-5
Poly(vinylidene fluoride-co-hexafluoropropylene)	Sigma-Aldrich	427179
PTFE Hydrophilic Membrane Filters, 0.22 mm, 90 mm	Scientificfilters.com/Tisch Scientific	SF13842
Critical commercial assays		
Folic Acid (FA) ELISA kit	Alpha Diagnostic International	Cat#0365-0B9
TMT10plex Isobaric Label Reagent Set	Thermo Fisher	A37725
Surface Plasmon Resonance CM5 kit	GE Healthcare	Cat#29104988
NanoBRET Target Engagement K-5 kit	Promega	Cat#N2500
Deposited data		
B-RAF RBD (apo) structure	This paper	PDB: 5J17
Structure of compound 5	This paper; Cambridge Crystallographic Data Center	CCDC: 2016466
Code for constraints-based modeling and analysis of autotrophic *E. coli*	This paper	https://gitlab.com/elad.noor/sloppy/tree/master/rubisco
Software and algorithms		

Gaussian09	Frish et al.[1]	https://gaussian.com
Python version 2.7	Python Software Foundation	https://www.python.org
ChemDraw Professional 18.0	PerkinElmer	https://www.perkinelmer.com/category/chemdraw
Weighted Maximal Information Component Analysis v0.9	Rau et al.[2]	https://github.com/ChristophRau/wMICA
Other		
DASGIP MX4/4 Gas Mixing Module for 4 Vessels with a Mass Flow Controller	Eppendorf	Cat#76DGMX44
Agilent 1200 series HPLC	Agilent Technologies	https://www.agilent.com/en/products/liquid-chromatography
PHI Quantera II XPS	ULVAC-PHI, Inc.	https://www.ulvac-phi.com/en/products/xps/phi-quantera-ii/